Five primitive-geometry counts and the Rudvalis correction

Leemans and Rodrigues [1] count primitive rank-two coset geometries up to automorphisms of their distinguished group, allowing the two types to be interchanged. Their Table 1 prints 1511 for the Rudvalis group. The accepted principal correction already disproves that number by the lower bound 50669. Here we obtain an exact total, conditional on the classical group classification and library inputs specified below. Four other entries are recovered by the same calculation.

Theorem 1. *Assume the named group and library premises stated below. For the groups* M_{11}, M_{23}, M_{24}, J_1 *and* Ru, *the numbers of firm, connected, flag-transitive primitive rank-two coset geometries, under the source's specified equivalence, are respectively*

$$37, \quad 170, \quad 5074, \quad 669, \quad 556287.$$

Proof. All five groups are nonabelian simple and have trivial outer automorphism group. These classical group facts are retained from the Atlas [2]. A maximal subgroup M is self-normalizing: its normalizer in G is either M or G, and the latter contradicts simplicity. Thus the conjugation action on all maximal subgroups is the disjoint union of the actions on G/M, taking one M from each conjugacy class. The source's geometries correspond exactly to the orbits on two-element subsets of this disjoint union. The full geometric and equivalence arguments are given in the previously accepted principal correction; the companion common proof states the same reduction explicitly.

Write ψ for the fixed-point function of this union and set

$$d = \psi(1), \quad s = \sum_{[g]} \frac{\psi(g)}{|C_G(g)|}, \quad r = \sum_{[g]} \frac{\psi(g)^2}{|C_G(g)|}, \quad u = \sum_{[g]} \frac{\psi(g^2)}{|C_G(g)|}.$$

The common proof derives each coset character from the subgroup and parent centralizer orders and the compatible fusion of classes. Its Theorem 2 then gives $(r - 2s + u)/2$ for the required unordered-pair count. No orbit representative or target count enters that calculation.

The retained library export contains all these integer inputs. The independent Python program evaluates the formulas using rational arithmetic, checks integrality of every fixed-point value, and obtains

G	s	d	r	u	$(r-2s+u)/2$
M_{11}	5	309	67	17	37
M_{23}	7	44414	316	38	170
M_{24}	9	1507537	9851	315	5074
J_1	7	13016	1230	122	669
Ru	15	398283420	1107296	5308	556287

The accompanying report gives every maximal-subgroup permutation character and all scalar products, so the displayed totals can be recomputed directly from the supplied inputs. The identity centralizers, class-size sums, subgroup indices, centralizer inclusions and fusion ranges are checked. Each coset character has orbit average one and values between zero and its degree. For every element class, the fixed unordered-pair count is a nonnegative integer. These checks and the exact final divisions establish the displayed arithmetic. □

Data premises and attribution

The input is the installed GAP Character Table Library 1.3.11 (25 May 2025), read by GAP 4.15.1. Its documented `Maxes` contract supplies one representative of every conjugacy class of maximal subgroups, their ordinary character tables, and compatible class fusions to the parent. The complete lists, the validity of the named tables, the stored square maps and the compatible fusions are external classification/data premises. The checks above do not prove those premises from internal numerical consistency. The exact library files, exporter and execution receipt are bound separately. The export terminated normally in less than one second; its retained output is 10788 bytes, with no standard-error output.

The second program does not call GAP: it derives the characters and averages independently from that export. Its five damage controls reject altered identity centralizers, out-of-range fusion and square maps, a duplicate maximal identifier, and a changed parent order. These are tests of the certificate interface, not substitutes for the general proof. The raw earlier aggregate probe is not needed as a premise or as an expected answer.

Burnside's lemma, induction of the trivial character, and the elementary coset realization are classical. The source supplies the problem and its equivalence convention. Our contribution is this uniform application, the exact independent arithmetic, and the correction of the Rudvalis entry under that convention. We make no claim that these formulas or the four agreeing totals are new.

The already established M_{22} total and remaining claims

The complete corrected L073 application proves that M_{22} has 81 pair classes under precisely the equivalence used here: an automorphism of the distinguished group and interchange of types. Its common Lemma 1 and Theorem 2 identify these pairs with the firm, connected primitive geometries, and its application Theorem 4 gives the exhaustive 81-class result. Thus the M_{22} entry of the present source is imported with its existing named-group, catalogue and exact algorithm premises. Its enumeration is not rerun and earns no new original-manuscript or measurement credit.

The other six outer-automorphism cases, all five printed suborbit tables, and the individual M_{11} incidence and design tables remain separate obligations. In particular, this calculation alone does not complete L036. It proves no recorded open conjecture, supplies no Lean certificate, and asserts no compression factor.

References

[1] D. Leemans and B. G. Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. **2** (2019), P1.08, doi:10.26493/2590-9770.1301.165. Table 1 is on printed page 5.

[2] J. H. Conway et al., *Atlas of Finite Groups*, Oxford University Press, 1985. Named sporadic groups and their outer automorphism groups are retained inputs.

[3] T. Breuer, *The GAP Character Table Library*, version 1.3.11, manual Section 3.7-1, `Maxes`. The installed package, documentation and selected data are bound in the request.